INCH-POUND

MIL-H-83534(USAF)
AMENDMENT 2
24 AUGUST 1995
SUPERSEDING
AMENDMENT 1
25 OCTOBER 1993

MILITARY SPECIFICATION

HEADSET-MICROPHONE, MONAURAL, LIGHTWEIGHT, INTRA-AURAL
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-H-83534, dated 1 September 1989, and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

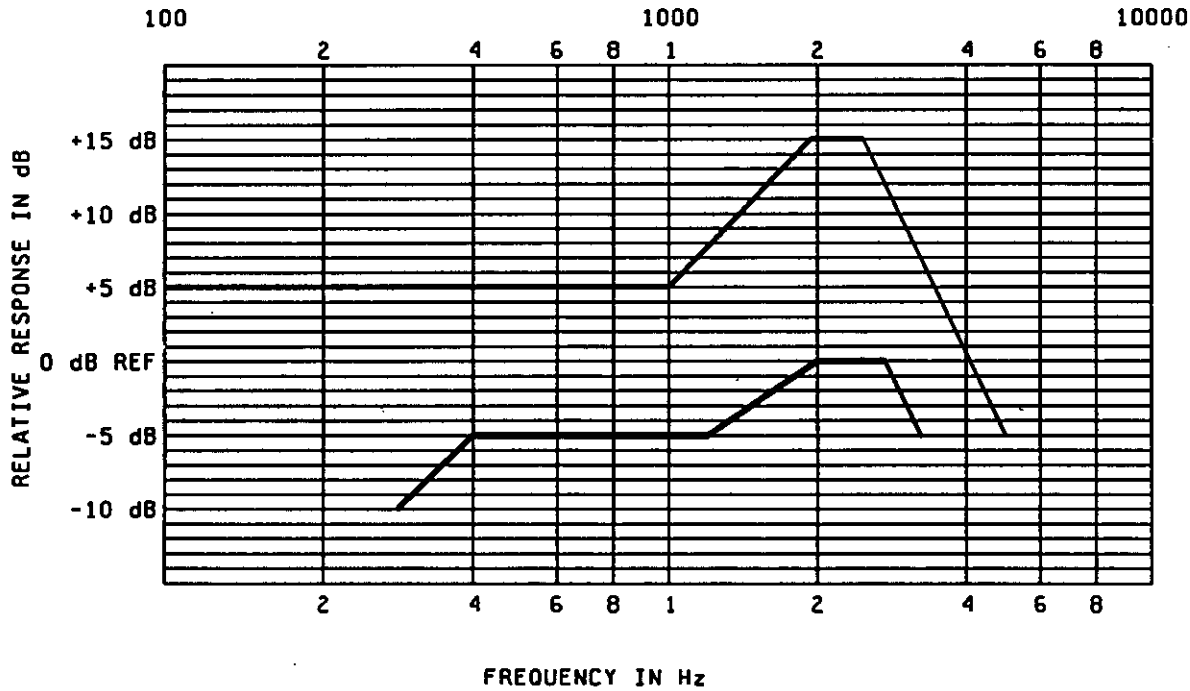
PAGE 11

FIGURE 7, capsule cable: Delete "40.0 ±2.0" inches and substitute "44.0 ±6.0" inches.

PAGE 20

FIGURE 15, delete and substitute:

"



NOTE: The response curve shown is typical and may 'float' within the envelope.

FIGURE 15. Transmit response envelope."

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PAGE 21

3.5.1, delete and substitute:

"3.5.1 Transmit output level. With the exception of test condition 3, when tested as specified in 4.8.2, the 1,000 Hz output level shall be -22.5 ±4 dBV. While under test condition 3, the 1,000 Hz output shall be within 6 dB of the level measured in test condition 4. When tested with an electrically similar headset connected in parallel, the 1,000 Hz output level shall be within 6.5 dB of the value measured as a single headset."

3.5.2, delete in its entirety.

PAGE 22

3.6.2, delete in its entirety.

3.6.7, delete and substitute:

"3.6.7 Receiver linearity. When a headset with volume control is specified (see 3.1), the receiver output shall remain within the envelope limits of figure 18 when tested as specified in 4.8.17."

3.6.8, delete and substitute:

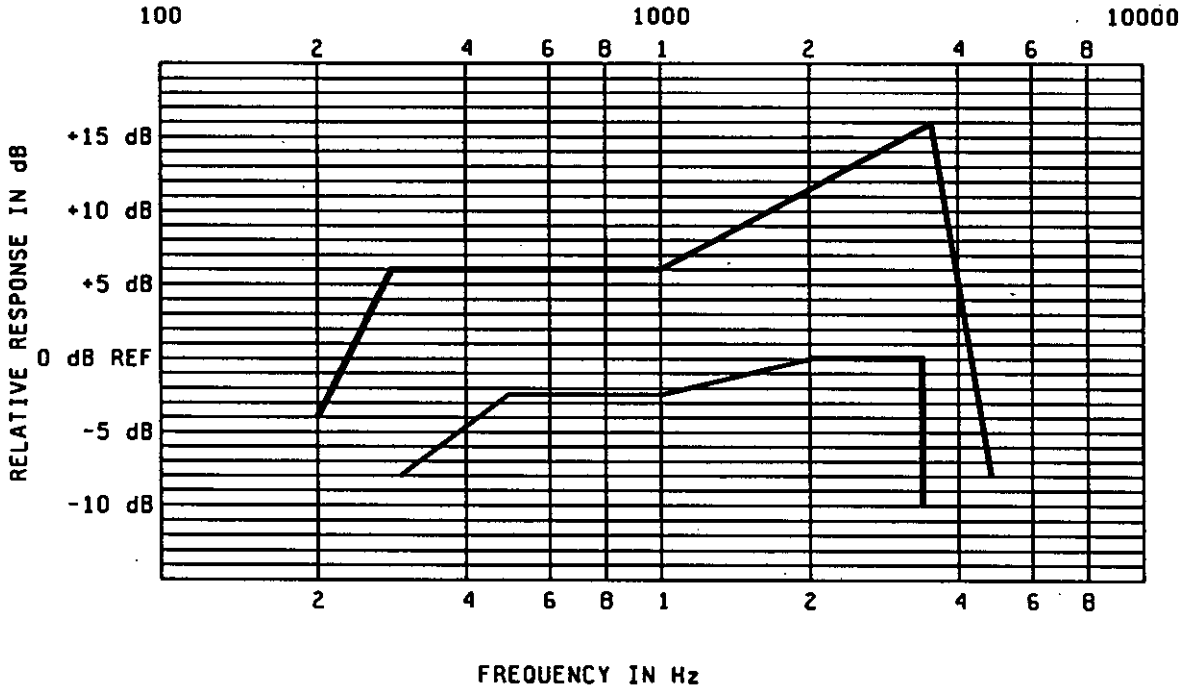
"3.6.8 Common mode rejection. When a headset with volume control is specified (see 3.1), the receiver output shall be at least 40 dB below the level of 3.6.4 when tested as specified in 4.8.18."

3.6.10, delete and substitute:

"3.6.10 Transition times. When a headset with volume control is specified (see 3.1), the transition time requirements of 3.6.10.1 through 3.6.10.4 shall apply."

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FIGURE 16, delete and substitute:
"



NOTE: The response curve shown is typical and may "float" within the envelope.

FIGURE 16. Receive response envelope, intra-aural headset with and without volume control."

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PAGE 27

* 4.2.b, delete "Qualification" and replace with "First Article".

* 4.6, title and first sentence, delete "Qualification" and replace both with "First Article". First sentence delete "(see 6.2)" and substitute "(see 6.5)".

* 4.6.1, first sentence, delete "Qualification" and replace with "First Article".

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* 4.6.3, first sentence, delete "Qualification" and replace with "First Article".

* 4.6.4, delete in its entirety.

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* TABLE II, title, delete "Qualification" and replace with "First Article".

TABLE II, Group I, Loudness level (TXLL): Delete.

TABLE II, Group I, Loudness level (RXLL): Delete.

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TABLE III, Loudness level (TXLL): Delete.

TABLE III, Loudness level (RXLL): Delete.

PAGE 31

4.7.2.1.1, delete and substitute:

"4.7.2.1.1 Sampling plan. Four sample units shall be selected from each 6-months production or each 2,500 units, whichever occurs first after date of notification of qualification."

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4.8.3, delete in its entirety.

PAGE 35

4.8.8, delete and substitute:

"4.8.8 DC resistance (see 3.5.7). The dc resistance in ohms shall be determined as V_h (in volts)/I_h (in amperes) (see figure 19)."

4.8.12, delete in its entirety.

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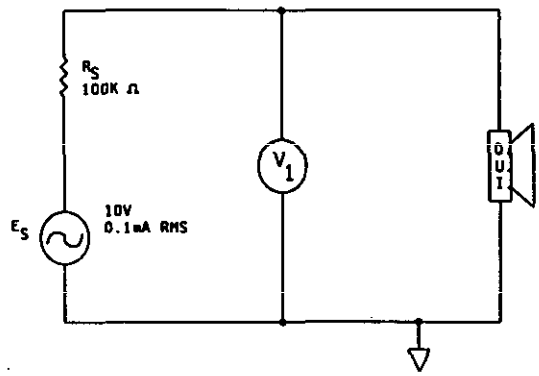
PAGE 38

4.8.21, delete and substitute:

"4.8.21 Input impedance. Receiver impedance shall be determined to be the average of measurements taken at 500 Hz, 1,000 Hz, and 2,000 Hz. Each measurement shall be taken using the constant current method described in figure 22."

$$I = \frac{10V}{100K\Omega} = 100uA$$

$$Z = \frac{V_1}{100uA} = 10^4\ V_1$$



NOTES:
1. Constant current generator may be made with oscillator in series with 100 ohm resistor. Other values of E_s may be used (3.0 V_{RMS} min.). Then

$$Z = \frac{V_1}{E_s} \times 100K\Omega$$

2. Do not use ammeter to set current. Use floating voltmeter (true RMS) across R_S to avoid ground loops
3. Example calculation: V_1 is 30 mV for 300 ohms Z and V_1 is 600 ohms Z.
4. All tolerances are ±1%.

FIGURE 22. Input impedance test circuitry. "

PAGE 39

4.8.26.1.b(3), delete in its entirety.

4.8.26.3, delete and substitute:

"4.8.26.3 Vibration (see 3.11.3). The headset shall be exposed to 0.5g peak (displacement less than .03 inch) sinusoidal vibration along each of three orthogonal axes. Frequency shall be swept logarithmically at 0.1 octave per minute from 5 to 100 Hz and 0.25 octave per minute from 100 to 500 Hz for a total of approximately 1 hour in each axis."

PAGE 39

4.8.27, delete and substitute:

"4.8.27 Cable flexing (see 3.12). The cable shall be run through 50,000 flex cycles over a 1/4-inch radius, 60° to each side of vertical at a rate of 85 flex cycles per minute with a load of 12 ±1/4 ounce attached to the lower end of the cable. One cycle equals two complete 120° arcs of travel. Failure of the cable shall be considered as a 25 percent increase in the initial resistance of the test sample in which the four or six conductors are connected in series."

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* 6.5, delete and substitute:

"6.5 First Article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article sample should consist of 6 units. The contracting officer should include the specific instructions in acquisition documents regarding arrangements for examinations and test approval of the documents first article."

NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

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